A
Technical Seminar Report
On

"TECHNICAL STUDY ON CONSTRUCTION AND PERFORMANCE OF MICRO PILES"

Submitted in partial fulfilment of the academic requirement for the award of the degree of

BACHELOR OF TECHNOLOGY
in
CIVIL ENGINEERING

From
JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
By

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Under the esteemed guidance of
Mr.G.NARENDER M.Tech
Assistant Professor

DEPARTMENT OF CIVIL ENGINEERING

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CERTIFICATE

This is to certify that the Technical seminar report entitled "**TECHNICAL STUDY ON CONSTRUCTION AND PERFORMANCE OF MICRO PILES**" is a bonafide record of the work done

By

M .SRIKANTH 20C15A0130

In the partial fulfilment for the award of degree of **Bachelor of Technology** in **CIVIL ENGINEERING** from **Jawaharlal Nehru Technological University Hyderabad, Hyderabad** during the Academic Year 2022- 2023.

Mr.G.NARENDER M.Tech
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DECLARATION

This is to certify that the Technical Seminar work Reported In the present title as "**TECHNICAL STUDY ON CONSTRUCTION AND PERFORMANCE OF MICRO PILES**" is a record of work done by me in partial fulfilment of the requirement for the award degree of Bachelor of Technology in Civil Engineering during academic year 2022-2023 sessions at the **ANURAG ENGINEERING COLLEGE,** Ananthagiri (V&M), affiliated to JNTU Hyderabad. To the best of our knowledge, the results contained in this thesis have not been submitted to any other university or institution for the award of any degree.

In all sincerely,

M.SRIKANTH (20C15A0130)

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ABSTRACT

Micro piles have been used effectively in many applications of ground improvement to increase the bearing capacity and reduce the settlements particularly in strengthening the existing foundations. Advantages of Micro piles are high carrying capacity, less site constraint problem and self sustained operation. This piling system is therefore attractive to both the client and foundation designer. Apart from this the light and compact drilling rigs other ancillary equipment, like grout mixer and grout pumps, is very compact size. Micro installations can penetrate to hundreds of feet in depth; each of the piles can support many tons of loads. Micro piles transfer loads through liquefiable soil to competent bearing strata to conform to design requirements. This paper deals with a case study in which micro piles of 100 mm diameter and 4 m long have been used to improve the bearing capacity of foundation soil and in the rehabilitation of the total building foundation system. The micro piles were inserted around the individual footings at inclination of 70° with the horizontal.Micropile technology is a reliable pile system that can withstand large capacity axial or lateral loads with minimal disturbance to the existing structures. They became very popular due to their ability to transfer loads efficiently through skin friction and due to their many installations advantages over conventional pile system. Micro piles has the capability of combining micro pile technology with one or more of the other ground improvement techniques to meet unique or complex project requirements cost effectively and efficiently.

Keywords: micro-piles, rehabilitation, ancillary equipment, liquefiable, reliable pile system.

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CHAPTER I

INTRODUCTION

The use of micropiles has grown significantly since their conception in the 1950s, and in particular since the mid-1980s. Micropiles have been used mainly as foundation support elements to resist static and seismic loads, and to a lesser extent, as in-situ reinforcements to provide stabilization of slopes and excavations. Many of these applications are for transportation structures.

1.1 MICROPILE DEFINITION AND DESCRIPTION

A micropile is a small-diameter (typically less than 300 mm (12 in.)), drilled and grouted non-displacement pile that is typically reinforced. A micropile is constructed by drilling a borehole, placing steel reinforcement, and grouting the hole as illustrated in Figure 1-1. Micropiles can withstand relatively significant axial loads and moderate lateral loads, and may be considered a substitute for conventional driven piles or drilled shafts or as one component in a composite soil/pile mass, depending upon the design concept employed. Micropiles are installed by methods that cause minimal disturbance to adjacent structures, soil, and the environment. They can be installed where access is restrictive and in all soil types and ground conditions. Micropiles can be installed at any angle below the horizontal using the same type of equipment used for the installation of ground anchors and for grouting projects. Since the installation procedure causes minimal vibration and noise and can be used in conditions of low headroom, micropiles are often used to underpin existing structures. Specialized drilling equipment is often required to install the micropiles from within existing basement or other limited headroom facilities. Most of the applied load on conventional cast-in-place drilled or non-displacement piles is structurally resisted by the reinforced concrete; increased structural capacity is achieved by increased cross-sectional and surface areas. Micropile structural capacities, by comparison, rely on high-capacity steel elements to resist most or the entire applied load.

These steel elements may occupy as much as one-half of the drillhole cross section. The special drilling and grouting methods used in micropile installation allow for high grout/ground bond values along the grout/ground interface. The grout transfers the load through friction from the reinforcement to the ground in the micropile

bond zone in a manner similar to that of ground anchors. Due to the small pile diameter, any end-bearing contribution in micropiles is generally neglected.

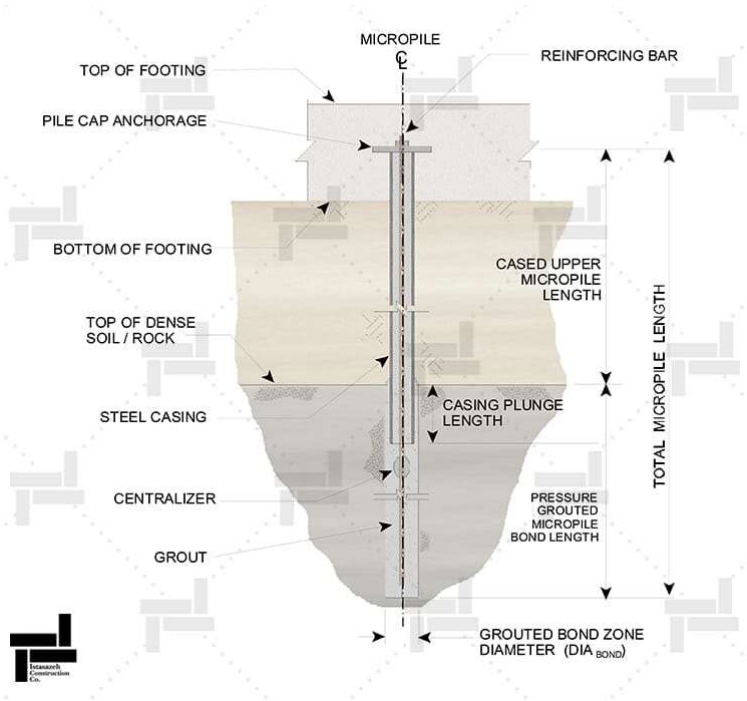


Fig1.1: Component parts of micro piles

1.2 HISTORICAL BACKGROUND

Micropiles were conceived in Italy in the early 1950s, in response to the demand for innovative techniques for underpinning historic buildings and monuments that had sustained damage with time, and especially during World War II. A reliable underpinning system was required to support structural loads with minimal movement and for installation in access-restrictive environments with minimal disturbance to the existing structure. An Italian specialty contractor called Fondedile, for whom Dr. Fernando Lizzi was the technical director, developed the palo radice, or root pile, for underpinning applications. The palo radice is a small-diameter, drilled, cast-in-place, lightly reinforced, grouted pile. Although steel was in short supply in postwar Europe, labor was inexpensive, abundant, and often of high mechanical ability. Such conditions encouraged the development of these lightly reinforced, cast-in-place root pile elements, largely designed and installed by specialty contractors on a design-build basis. Load testing on these new root piles measured capacities in excess of 400 kN (90 kips), although the design capacity based on contemporary conventional bored

pile design methodologies suggested capacities of less than 100 kN (23 kips). Direct full-scale load tests were performed at relatively little cost, fostering the acquisition and publication of a wealth of testing information. No grout/ground bond failures were recorded during these early tests. The use of root piles grew in Italy throughout the 1950s. Fondedile introduced the technology in the United Kingdom in 1962 for the underpinning of several historic structures, and by 1965, it was being used in Germany on underground urban transportation projects. For proprietary reasons, the term “micropile” replaced “root pile” at that time.

CHAPTER II

LITERATURE REVIEW

Micro pile :Recent Advances and Future Trends by Dr. Jaykumar Pitroda, Jaydev Jagmohandas Bhavsar

The conceptual idea behind this important technological development was to create a type of pile that would be able to carry large loads while causing minimal vibration or disturbance to in situ materials at the time of installation. The rigs required to install them are often relatively small. Because of these important advantages, micro piles have been widely used in seismic retrofitting, in the rehabilitation of foundations of structures that are very sensitive, and in locations with low headroom and severely restricted access conditions. Furthermore, micropiles have been increasingly used, not only as underpinning foundation elements, but also as foundations for new structures.

A Model Study of Micro piles subjected to Lateral Loading and Oblique Pull by Binu Sharma As structural support it can be used for underpinning of distressed historical monuments, seismic retrofit mainly in congested and low headroom areas, resisting uplift dynamic loads. As in situ reinforcement it can be used for slope stabilization, for arresting structural settlement, excavation support in congested areas and as retaining structures. Its wide application makes it necessary to study the soil micropile interaction under various loading conditions.

Micropiles applications for siesmic retrofitting of historical bridges by Ferit Cakir and Jamshadi Mohammadi

Traditional retrofitting techniques have been inadequate for improving seismic behavior and resistance of these structures. With current advancement in materials and construction techniques, new technologies that can be appealing to historical bridges are emerging. Among these techniques include underpinning using micropiles with the technique called “micropiling”. Today, micropiles are used for the structural and seismic retrofitting of bridges, mosques, churches and many other ancient cultural heritage and modern structures.

CHAPTERS III

MICROPILE CLASSIFICATION SYSTEM

The design of an individual micropile or a group of micropiles differs greatly from that of a network of closely-spaced reticulated micropiles. This difference led to the definition of CASE 1 micropile elements, which are loaded directly and where the micropile reinforcement resists the majority of the applied load (Figure 2-1). CASE 2 micropile elements circumscribe and internally reinforce the soil to theoretically make a reinforced soil composite that resists applied loads (Figure 2-2). This is referred to as a reticulated micropile network. CASE 1 micropiles can be used as an alternative to more conventional types of piles since they are used to transfer structural loads to a deeper, more competent or stable stratum. Such directly loaded piles, whether for axial (compression or tension) or lateral loading conditions, are referred to as CASE 1 elements. The load is primarily resisted structurally by the steel reinforcement and geotechnically by side resistance developed over a “bond zone” of the individual micropiles. At least 90 percent of all international applications to date, and virtually all of the projects in North America have involved CASE 1 micropiles. Such micropiles are designed to act individually, although, they may be installed in groups. Typical arrangements of CASE 1 micropiles are illustrated in Figure 2-3.

The remaining micropile applications involve networks of reticulated micropiles as components of a reinforced soil mass which is used to provide stabilization and support. These micropiles are referred to as CASE 2 elements. Structural loads are applied to the entire reinforced soil mass, as compared to individual micropiles (as for CASE 1 micropiles). CASE 2 micropiles are lightly reinforced because they are not individually loaded. A conceptual application of a network of reticulated micropiles is illustrated .

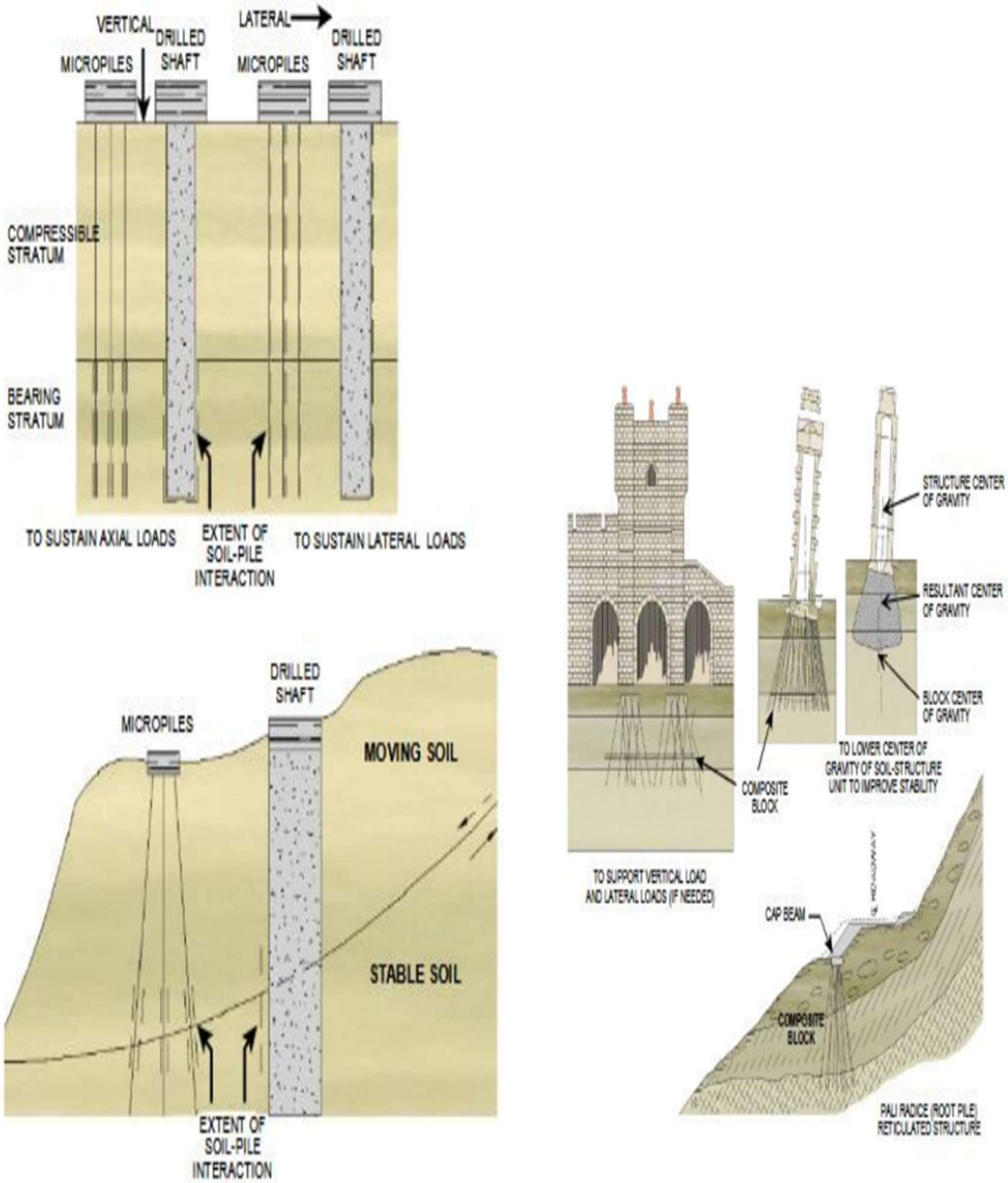


Fig 3.1 Case 1 and Case 2 Micro piles

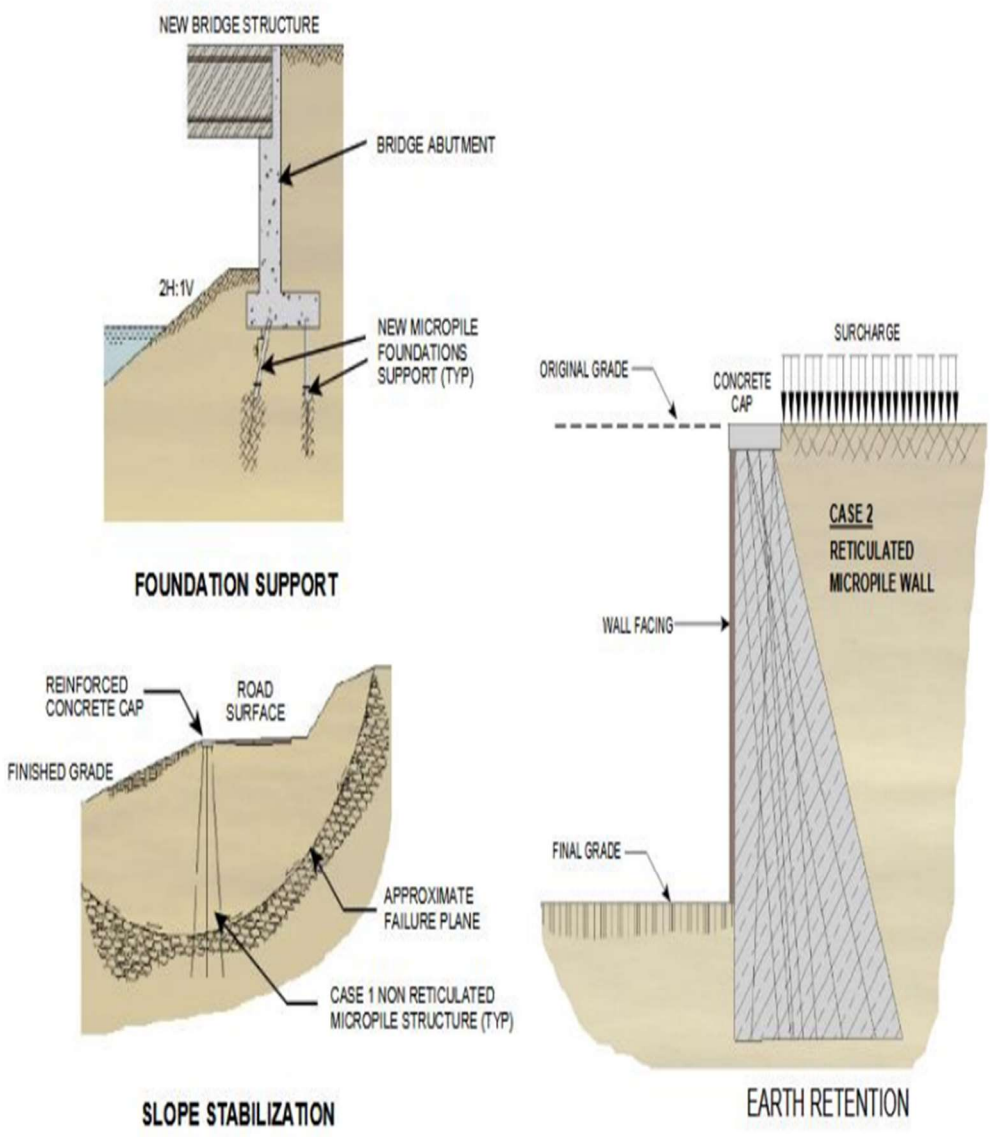


Fig 3.2 Case 1 and Case 2 Micro piles Arrangement

CHAPTER-IV

CONSTRUCTION TYPE CLASSIFICATION

The method of grouting is typically the most sensitive construction process influencing grout/ground bond capacity. Grout/ground bond capacity varies directly with the grouting method. The second part of the micropile classification consists of a letter designation (A through D) based primarily on the method of placement and pressure under which grouting is performed during construction. The use of drill casing and reinforcement define sub-classifications. The classification is shown schematically in Figure 2-5 and is described subsequently.

• **Type A:** For Type A micropiles, grout is placed under gravity head only. Sand-cement mortars or neat cement grouts can be used. The micropile excavation may be underreamed to increase tensile capacity, although this technique is not common or used with any other micropile type.

• **Type B:** Type B indicates that neat cement grout is placed into the hole under pressure as the temporary drill casing is withdrawn. Injection pressures typically range from 0.5 to 1 MPa (72 to 145 psi) to avoid hydrofracturing the surrounding ground or causing excessive grout takes, and to maintain a seal around the casing during its withdrawal, where possible.

• **Type C:** Type C indicates a two-step process of grouting including: (1) neat cement grout is placed under gravity head as with Type A; and (2) prior to hardening of the primary grout (after approximately 15 to 25 minutes), similar grout is injected one time via a sleeved grout pipe without the use of a packer (at the bond zone interface) at a pressure of at least 1 MPa (145 psi).

• **Type D:** Type D indicates a two-step process of grouting similar to Type C. With this method, neat cement grout is placed under gravity head (as with Types A and C) and may be pressurized (as for Type B). After hardening of the initially placed grout, additional grout is injected via a sleeved grout pipe at a pressure of 2 to 8 MPa (290 to 1,160 psi). A packer may be used inside the sleeved pipe so that specific horizons can be treated several times, if required.

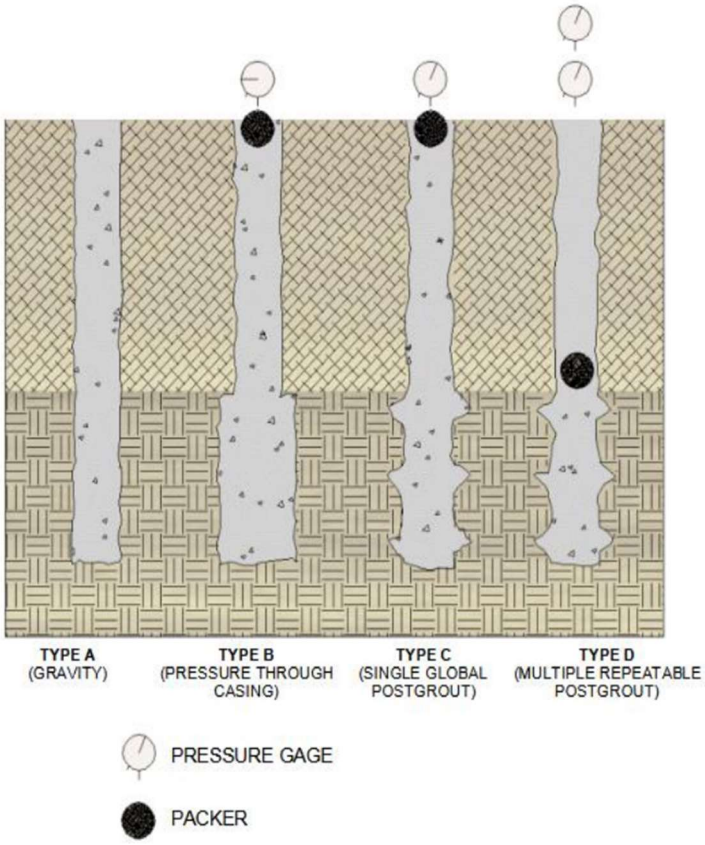


Fig 4.1 Micro Classification System Based On Type Of Grounting

4.1 MICROPILE APPLICATIONS IN TRANSPORTATION PROJECTS

Micropiles are currently used in two general application areas:

(1) structural support; and

(2) in-situ reinforcement .

Structural support includes new foundations, underpinning of existing foundations, seismic retrofitting applications and foundation support for earth retaining structures. Insitu reinforcement is used for slope stabilization and earth retention; ground strengthening; settlement reduction; and structural stability. Selection factors influencing the choice of micropiles for a project application are described herein. Example micropile applications.

CHAPTER V

CONSTRUCTION OF MICROPILES

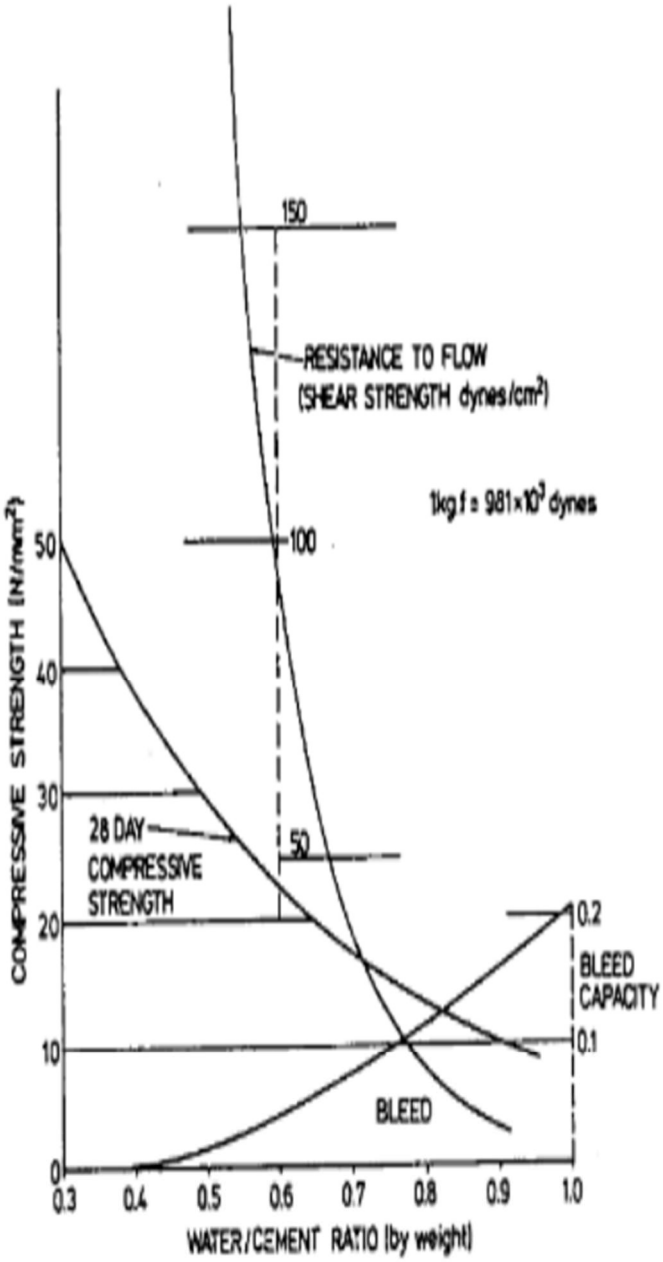
5.1 Drilling Techniques

The drilling method is selected on the basis of causing minimal disturbance. The ground and nearby sensitive structures and able to achieve the required drilling performance. In all drilling methods, drilling fluid is used as a coolant for the drill bit and as a flushing medium to remove the drill cuttings. Water is the most common drilling fluid compared to other drilling fluids such as drill slurries, polymer, foam and bentonite . Another type of flushing medium is using compressed air, which is commonly used in Malaysia.

Fig 5.1 Large Track mounted pile rig

Fig 5.2 Rotary Hydraulic Drill rig

Grouting Micro pile grouting equipment consists at a minimum of a colloidal high speed, high shear mixer, and holding tank grout pump capable of reaching pressures of 300psi, pressure gauges, and re-circulation lines. The colloidal mixer is a high-shear grout plant that is capable of rapidly mixing neat cement based grout in a few minutes, with a thorough wetting of the individual cement grains. A thorough wetting allows a low water-cement ratio grout to be pumped easily through the grout lines that run from the plant to the pile. Without a colloidal plant, clumps of cement will cling together, clogging injection lines, and ultimately yielding a lower strength grout, because significant amounts of the cement grains are not hydrated.



$1\ N/mm^2 = 145\ psi$

$1\ dyne/cm^2 = 1.45 \times 10^{-5}\ psi$

Fig 5.3 Effect of water content on grout compressive strength and flow properties

Following a thorough mixing, the grout must be stored in an agitation tank with agitation blades that constantly stir the mixed grout, prolonging separation of the cement from the mix water. With proper admixtures, grout life may be extended easily to a working time of 6 hours, and in some cases, may be suspended indefinitely until the reaction is reinitiated on demand. A Marsh Cones implifies optimization of the grout mix design with respect to retarders and fluidifiers. A Marsh Cone time of 11–14 seconds is typically considered ideal for pumping into micropile packer systems

5.2 Installation

The micro pile contractor must keep accurate and contemporaneous logs showing detailed installation information. Boring logs are used for planning the pile installation, but in karst' conditions, rockhead may vary greatly within a few feet of the boring. In karst,it is conceivable that each pile will be unique in overall length, un-bonded length, depth to rock head and sequencing of adequate bearing stratum. The pile installation log is the only record of the verification of the pile design dimensions and compositions having been achieved. The pile installation log also generates the pay items for the project, such as drilled footage in soil as well as rock, and amount of grout consumed. The contractor must be able to verify through his logs that the design requirement of a given amount of rock has been encountered in order to substantiate that a pile of the design capacity has been installed.

Single tube advancement: By this method, the toe of the drill casing is fitted with an open crown orbit, and the casing is advanced into the ground by rotation of the drillhead. Water flush is pumped continuously through the casing, which washes debris out and away from the crown. Rotary duplex: With the rotary duplex technique, drill rod with a suitable drill bit is placed inside the drill casing. It is attached to the same rotary head as the casing, allowing simultaneous rotation and advancement of the combined drill and casing string.

Rotary percussive duplex: Rotary percussive duplex systems are a development of rotary duplex methods, where by the drill rods and casings are simultaneously percussed, rotated, and advanced. The percussion is provided by a top-drive rotary percussive drillhead. This method requires a drill head of substantial rotary and percussive energy.

Fig 5.4 Drill rods and casing shoes

5.3 MICROPILE CONSTRUCTION MONITORING AND INSPECTION

Specifications describe minimum requirements for micropile materials. These minimum requirements may be defined explicitly or by reference to standard specifications such as those set forth by AASHTO and ASTM. Because the materials used to construct a micropile are common civil engineering materials, prequalified products are not commonly required, however, certain proprietary features may be involved in, for example, the micropile to footing connection. Recommendations for the storage and handling of selected materials are commonly included in specifications. Conformance to a specification requirement is commonly assessed in one of the following ways:

(1) reviewing manufacturer or supplier certification submitted by the contractor;

(2) reviewing of product literature and visual inspection; or

(3) conformance testing.

All materials and prefabricated elements delivered to the site must be visually examined prior to installation to verify required geometry and dimensions and to identify any defects in workmanship, contamination, or damage by handling. All nonconforming materials are unacceptable, unless appropriate corrections are made in accordance with specifications or by written approval of the project engineer.

5.4 Storing and Handling of Cement

Cement is typically supplied either in bagged (43 kg/sack) or bulk (1320 kg/bulk bag) form, depending on site conditions, job size, local availability, and cost. Cement should not be stored directly on the ground or anywhere where it may be subject to significant moisture absorption. The contractor should avoid stacking the bags too high to prevent over compaction of the cement. Cement should be visually examined for lumps or other indications of hydration or contamination by foreign matter immediately prior to its use. The water used for grout mixing must be potable, and not contain impurities harmful to the steel or the grout. The contractor should also handle admixtures with care following manufacturer's recommendations. Approved admixtures that have exceeded the manufacturer's recommended shelf life should not be used.

5.5 Storing and Handling of Reinforcing Steel

Steel reinforcement should be stored above the ground surface and be protected against mechanical damage and exposure to weather. Reinforcing bars should be lifted using fiber ropes or webbing and should be supported at several locations to prevent excessive bending. The micropile contractor should use care in unloading and storing all reinforcing steel material. The storage area should be away from construction traffic and in a protected location. When reinforcing steel material is delivered to the job site, it should be visually checked for size and any signs of corrosion. Properly stored steel reinforcing materials prevent corrosion or contamination by dirt, oils, and/or organics. Wood spacers placed between the ground and he steel materials will prevent slight rusting that can occur if steel is exposed to the ground. A light coating of surface rust on the steel is normal and indicates that oil and grease are not present. Deep flaky corrosion or deep pitting of the steel is cause for rejection. All steel materials should be inspected just prior to their use. Double corrosion-protected and/or epoxycoated steel bars should be delivered to the job site pre-manufactured. Extra care in handling and storage shall be given to these materials. Epoxy-coated bars shall be properly wrapped with padded bands and placed on wooden or padded supports. Premanufactured corrosion-protected steel bars must be stacked with care to prevent any damage to the corrugated tube and/or

epoxy coating. Any repairs to epoxy coatings shall be performed in accordance with manufacturer's recommendations.

Fig 5.5 Storage of reinforcing Steel

Fig 5.6 Double corrosion Protected Reinforcing steel

5.6 Inspection of Drilling Activities

The contractor's drilling operation should not cause unacceptable loss of ground. Signs of unacceptable loss of ground include the inability to withdraw the drill casing from the hole, large quantity of soil removal with little or no casing advancement, and subsidence of the ground above drilling location. The QA inspector should document this information and other observations of the drilling procedures in their daily reports. When temporary casing is to be removed during the grouting process, it should be removed in a manner that does not cause the pile reinforcement to become disturbed, damaged, or allowed to loss contact with the in-situ soil. The casing should also be kept full of grout during the removal operation to minimize the potential for drillhole collapse. The contractor is responsible for implementing all measures required to maintain an open drillhole during construction. Fluid levels in the drillhole must always be kept above the ground water level to minimize potential ground losses into the drillhole from potentially unstable soils. Drawings and Specifications should be recorded by the QA inspector and this information provided to the engineer.

5.7 Inspection of Reinforcement Installation Activities

Reinforcement may be placed either prior to drill hole grouting, or placed into the groutfilled borehole before temporary casing (if used) is withdrawn. In any case, the inspector shall record the total pile length and bond zone length for each installed micropile. The reinforcement shall be capable of being inserted to its prescribed length in the hole without force. Inability to achieve this penetration in uncased holes usually means caving of soil into the hole has occurred, and withdrawal of the reinforcement and re drilling is necessary. Driving the reinforcement to full length or cutting off the reinforcement shall not be allowed. Care should be exercised so as not to damage corrosion protection measures or centralizers during reinforcement installation. Neither epoxy-coated, double corrosion-protected, or uncoated steel reinforcements should be dragged across abrasive surfaces or through the surface soil. All splices and couplings should be checked for proper seating. The reinforcement must be clean

of deleterious substances such as surface soil, oil, and mud, which may contaminate the grout or coat the reinforcement. Centralizers and spacers shall be checked to ensure placement at specified intervals, typically 2.5 to 3 m (8 to 10 ft) Attachment of the centralizers to the reinforcement should also be checked to ensure that the reinforcement remains centered in the borehole. Typically centralizers are attached with tie-wire or duct tape. If tremie grout tubes are placed after the reinforcement has been positioned, care should be exercised so that the tremie tube does not interfere with the centralizers**.**

5.8 Inspection of Grouting Activities

Before any grouting begins, the condition of the grouting equipment should be checked by the contractor and the grout-mix design should be verified. Care in execution of the grouting process can greatly impact the quality of the completed micropile and its load carrying capacity. Prior to grouting, the inspector should verify the diameters of micropiles drilled without casings. Ideally, the contractor should grout the micropiles immediately after drilling the bond zone. However, if this is not possible, grouting should be performed before degradation of the drill hole. The primary grouting operation involves injecting cement grout at the lowest point of the borehole so the hole will fill evenly without air voids. A tremie tube is typically tied to the reinforcing steel before insertion in the borehole.

CHAPTER VI
PILE LOAD TESTING

Load tests on piles are conducted on completion of 28 days after casting of piles. Two types of tests namely initial and routine tests, for each type of loading viz. vertical, horizontal (lateral) pull out, are performed on piles.

6.1 Initial Load Tests on Piles

This test is performed to confirm the design load calculations and to provide guidelines for setting up the limits of acceptance for routine tests. It also gives an idea of the suitability of the piling system. Initial Test on piles are to be carried out at one or more locations depending on the number of piles required. Load applied for the initial (cyclic) load test is 2.5 times the safe carrying capacity of the pile. Loading for Initial Tests is conducted as per Appendix 'A' Clause 6.3 of IS-2911 Part IV.

6.2 Routine Load Tests on Piles

Selection of piles for the Routine Test is done based on number of piles required subject to maximum of ½% of total number of piles required. The number of tests may be increased to 2% depending on the nature / type of structure. The test load applied is 1½ times the safe carrying capacity of the pile. The Maintained load method as described in Clause 6.2 of IS-2911 (Part IV) – 1985 shall be followed for loading for the Routine Tests. This test will be performed for the following purposes: a) To ensure the safe load capacity of piles b) Detection of any unusual performance contrary to the findings of the Initial Test. The tests shall be performed at the cut-off level only. A detailed report for the test result is prepared.

6.3 Vertical Load Tests on Piles

Pile Head - The pile head shall be chipped off till sound concrete is met wherever applicable. The reinforcement shall be cut and head levelled with Plaster of Paris. A bearing plate with a hole shall be placed on the head for the jack to rest. Reaction- Kentledge shall be suitably designed to get the desired reaction on the piles. Anchor piles (if required) shall be placed at a centre to centre distance of 3 times the pile diameter subject to a minimum distance of 2 M. Settlement- 2 dial gauges for

a single pile and 4 dial gauges for a group of piles with 0.01 mm sensitivity shall be used. They shall be positioned at equal distance around the piles on datum bars resting on immovable supports at a distance of 3D (min. of 1.5 m) where D is the diameter of pile or circumscribing circle for non-circular piles. Application of load- It shall be applied as specified depending on the type of test (routine / initial). Each load shall be maintained till the rate of displacement of the pile top is either 0.1 mm in the first 30 minutes or 0.2 mm in the first one hour or 2 hours whichever occurs first. The next increment in the load shall be applied on achieving the aforesaid criterion. The test load shall be maintained for 24 hours. Initial Tests- The safe load on a single pile shall be the least of the following: (i) 2/3rd of the final load at which the total displacement attains a value of 12 mm unless otherwise required in a given case on the basis of nature and type of structure in which case, the safe load should be corresponding to the stated total displacement permissible

6.4 Lateral Load Tests on Piles

The jack should be placed horizontally, between two piles. The load on the jack shall be the same on both the piles. The load will be applied in increments of 20% of the estimated safe load and at the cut off level. The load will be increased after the rate of displacement is nearer to 0.1 mm per 30 minutes. If the cut-off level is approachable, one dial gauge exactly at the cut-off level shall measure the displacement. In case the cut-off level is not approachable, 2 dial gauges 30 cm apart vertically, shall be set up and the lateral displacement of the cut-off level calculated by similar triangles. The safe load on the pile shall be the least of the following: a) 50% of the final load at which the total displacement increases to 12 mm. b) Final load at which the total displacement corresponds to 5 mm.

6.5 Pull out Tests on Piles

A suitable set up shall be designed to provide an uplift force to the piles. The load increments and the consequent displacements shall be as per the case of a vertical load test.

Fig 6.1: Pull-out test on piles

The safe load shall be the least of the following:

a) 2/3rd of the load at which the total displacement is 12 mm or the load corresponding to a specified permissible lift.

b) Half of the load at which the load displacement curve shows a clear break.

CHAPTER VII

ADVANTAGES AND DISADVANTAGES,GRAPHS

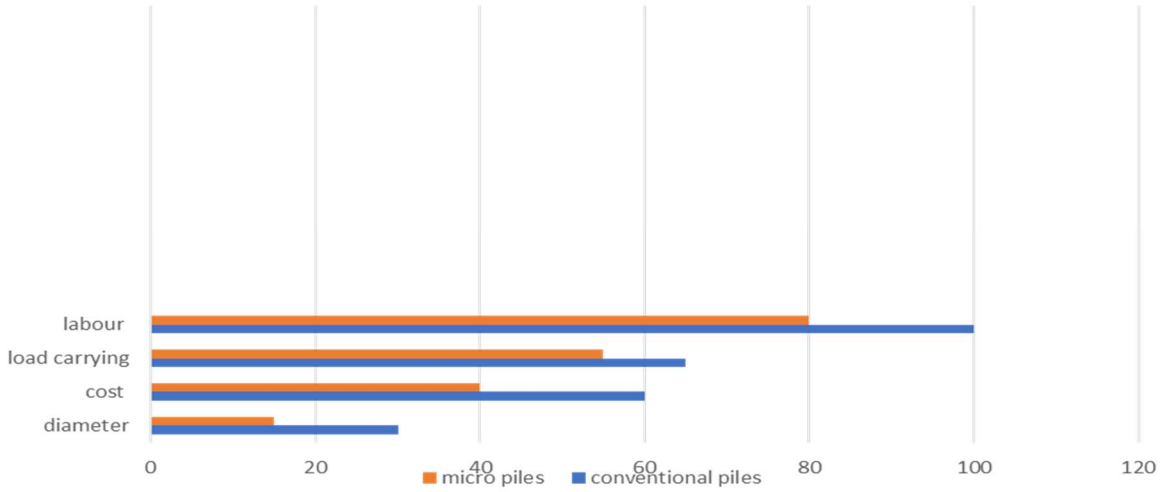
7.1 ADVANTAGES

1. Micropiles are used to reduce the existing structure where the requirement of minimal vibration or noise is of major importance.
2. They might be simply positioned where low headroom is difficult.
3. Micro piles don't require a big entry street or drilling platform.
4. They might be simply installed at any angle horizontally using floor anchors and similar tools used for floor anchors and grouting project.

7.2 DISADVANTAGES

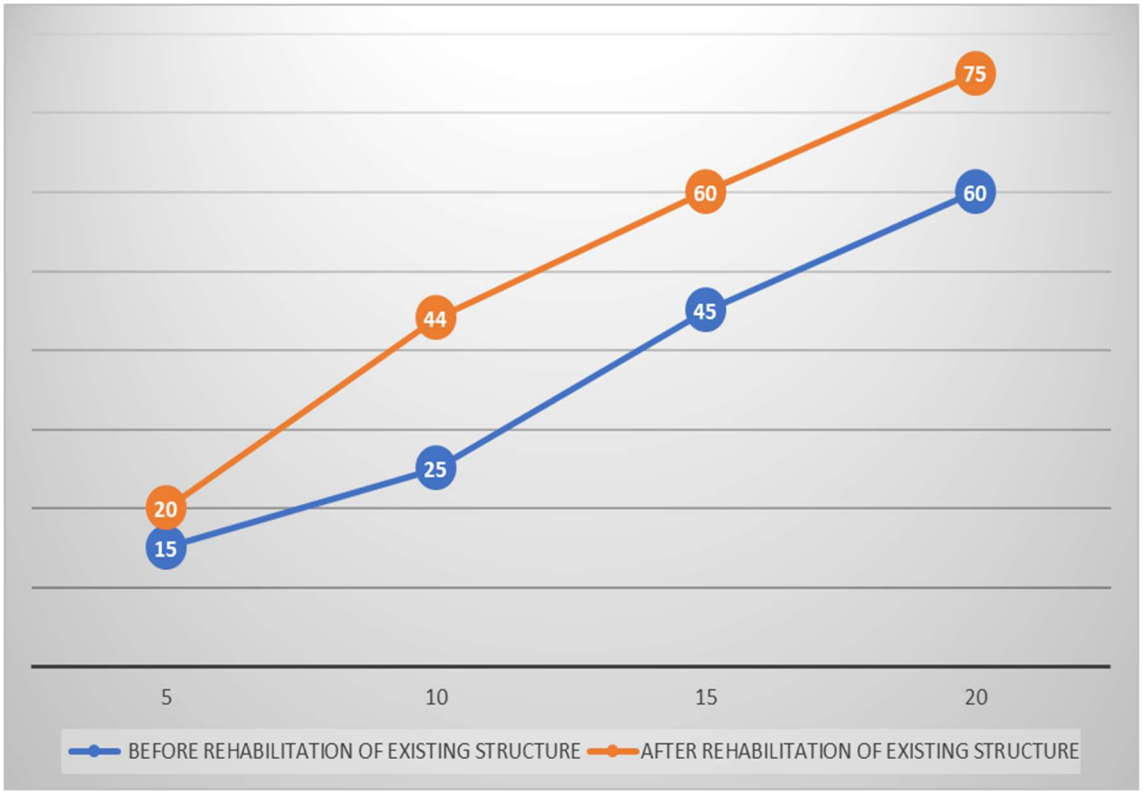
1. Higher cost as compared to other pilling systems.
2. Low lateral stiffness and flexural rigidity due to the small diameter
3. This limitations can be handled in current practice ,by installing the micropiles with inclined condition or providing a steel casing

7.3 WHY MICROPILES ARE PREFERRED OVER CONVENTIONAL PILES



Graph 1 MICROPILES ARE PREFERRED OVER CONVENTIONAL PILES

7.4 COMPRESSION OF EXISTING STRUCTURE BEFORE AND AFTER REHABILITATION



Graph 2 AGE OF EXISTING STRUCTURE

1. In conventional piles heavy machinery like hydraulic rigs and cranes required And micropiles need compact rotary drills and simple grouting machine

2. Piles gets abondonded when encountered with premature refusal due to intermediate obstruction conversation piles

3. Micropiles can penetrate on any surface like rock, existing foundation,soil where there is flooring boulders etc...

CHAPTER VIII
CONCLUSION

The following conclusion comes through the study of Micropile technology is gaining rapid acceptance. Although significant advances continue to come as a product of contractor innovation, significant efforts have been expanded and are underway by different entities to accommodate the ever changing applications of micropiles. Advances have been realized in the form of increased capacities, and introduction of new materials and design methods for integrating micropiles into existing projects. As demonstrated by FOREVER, JAMP, and numerous other commercial projects completed to date, micropiles provide viable solutions to difficult problems. Although sometimes designers wish to accommodate every conceivable loading condition, it is usually more appropriate to follow design procedures that balance safety and economic concerns. Further research is necessary on several aspects of micropile behavior. The success of these efforts will be fundamental for future development of micropile technology and to create new areas of application. Micropile technique is now widely used in rehabilitation works and is likely to be used even more in the future for strengthening and foundations.

CHAPTER IX

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